

Plant Disease Detection

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Abstract— This project aims to improve the robustness and edge-deployability of a lightweight plant leaf disease classifier. We reproduce the model presented in *Frontiers in Plant Science* (2023), where MobileNetV3-Small was fine-tuned on the PlantVillage dataset and later compressed via post-training quantization into an ONNX INT8 model. The original work reported ~99.5% of test accuracy and a reduction from ~1.5M to 0.93M parameters without accuracy loss, demonstrating suitability for edge devices.

PlantVillage images are captured in controlled laboratory settings and do not reflect real field conditions. Prior studies show significant performance degradation on real-world images containing shadows, cluttered backgrounds, and uneven lighting. To address this, we aim to improve generalization while respecting edge constraints. Our experimental plan includes reproducing the baseline with leak-proof splits, evaluating out-of-distribution performance on PlantDoc, applying domain-robust augmentations, improving probability calibration using temperature scaling, and comparing post-training quantization with quantization training. We expect improved real-world accuracy, better calibration, and preserved performance after compression, yielding a more reliable edge-ready plant disease detector.

The full implementation is available at:
<https://github.com/devasenan134/mobile-leafdoc.git>

Keywords—Deep Learning, MobileNetV3, Plant Disease Classification, Edge AI, Quantization, Transfer Learning, Domain Generalization, PlantVillage, PlantDoc.

I. INTRODUCTION

Plant diseases pose a significant threat to global food production, and early visual diagnosis remains one of the most accessible tools for farmers and agronomists. Recent advances in deep learning, particularly convolutional neural networks (CNNs), have enabled highly accurate image-based disease classification systems. However, many of these systems are trained on laboratory-style datasets such as PlantVillage, where leaves are captured under uniform lighting, plain backgrounds, and controlled imaging conditions. As a result, models trained exclusively on such datasets achieve near-perfect accuracy in-distribution but often fail when deployed in real agricultural environments, where images contain shadows, cluttered backgrounds, occlusions, and inconsistent lighting.

The paper Plant Disease Detection Model for Edge Computing Devices (*Frontiers in Plant Science*, 2023) demonstrated that a lightweight CNN **MobileNetV3-Small** can achieve ~99.5% test accuracy on PlantVillage and can be compressed through post-training quantization (INT8) to reduce parameters from ~1.5M to 0.93M with negligible loss in accuracy. This makes the model attractive for edge devices such as mobile phones, drones, Raspberry Pi units, and embedded systems where memory and compute resources are limited. However, the original work does not evaluate performance under domain shift, leaving open questions about real-world reliability.

To address this gap, our project focuses on reproducing the baseline model and extending it to improve performance in real-world scenarios. We incorporate the PlantDoc dataset, an out-of-distribution (OOD) dataset composed of mobile and online images captured in natural field conditions to evaluate the generalization ability of the model. Initial zero-shot experiments reveal a drastic drop in accuracy, confirming the sensitivity of lightweight CNNs to domain mismatch.

To improve robustness while maintaining edge constraints, we explore strategies including (1) leak-proof and stratified dataset splits to avoid overfitting, (2) domain-robust augmentations and color constancy in the training pipeline, (3) probability calibration using temperature scaling, and (4) a comparison of post-training quantization (PQT) versus quantization-aware training (QAT). Our goal is to develop an “edge-ready” plant disease detector that maintains high accuracy, strong calibration, and efficient on-device performance even when confronted with real-world, noisy images.

II. RELATED WORK

A. Deep Learning for Plant Disease Detection

Deep learning has become a popular approach for plant disease detection because it can automatically learn leaf patterns and disease features directly from images. These models perform extremely well on clean, lab-style datasets, but their accuracy often drops with real-world images with noise, shadows, and background clutter.

B. Lightweight CNN Architectures

Lightweight neural networks are designed to run efficiently on mobile and edge devices by reducing parameters and computational cost. They make real-time plant disease

detection more practical in the field, but may be more sensitive to noisy inputs compared to larger models.

C. Domain Shift in Agricultural Imaging

Plant disease datasets collected in labs differ from real-world field images, leading to a domain shift. Models trained on clean backgrounds and ideal lighting often struggle when tested on natural images with clutter, shadows, and inconsistent leaf conditions.

D. Edge AI and Model Compression

Edge AI focuses on running models directly on resource-limited devices such as phones or microcontrollers. Compression methods like pruning and quantization reduce model size and speed up inference, making deployment feasible without cloud dependence.

E. Quantization and Calibration Techniques

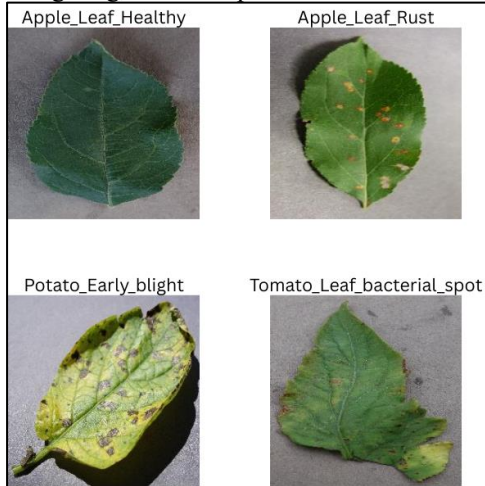
Quantization reduces model precision (e.g., FP32 to INT8) to improve speed and memory efficiency on edge hardware. Calibration techniques like temperature scaling help ensure that model output probabilities remain reliable after compression.

III. METHODOLOGY/TECHNICAL APPROACH

A. Datasets

1. PlantVillage (In-Distribution)

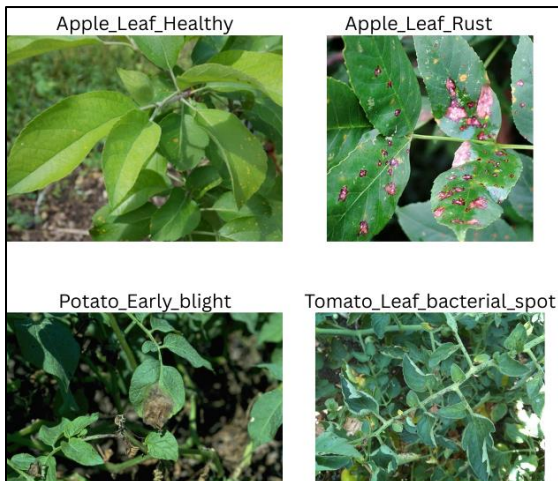
A large, curated dataset with uniform backgrounds and controlled lighting, used to reproduce the baseline model.



Samples from the PlantVillage dataset

2. PlantDoc (Out-of-Distribution)

A smaller field dataset containing natural lighting, shadows, and clutter, used to measure real-world performance.



Samples from the PlantDoc dataset

3. Unified Label Mapping

Both datasets use different naming conventions, so overlapping classes are mapped to a single consistent label space.

4. Preprocessing

All images are resized, and split using stratified, leak-proof methods to ensure fair evaluation.

B. Baseline Model

To have an in-distribution reference by reproducing the base paper's MobileNetV3-Small model on the PlantVillage(PV) dataset. The MobileNetV3 model uses a compact convolutional design optimized for low computational cost and fast inference on edge devices. Its lightweight structure allows efficient processing while maintaining strong feature extraction capabilities.

Training Configuration: Training and testing are solely based on the PV dataset (with no PlantDoc exposure whatsoever). To ensure data distribution remains unchanged, we apply a very moderate augmentation regime, use the same transforms for evaluation across different runs, and report top-1 accuracy along with macro-F1. Random seed and stratification are kept constant in all experiments to make the comparisons valid.

Outcome: The reproduced baseline reaches 96% top-1 accuracy on PV test—about +1 percentage point higher than the Frontiers-2023 number. The learning curves are regular, and the remaining errors are few and usually the case of borderline examples. This supports (i) that the backbone and optimization are correctly set up, and (ii) that PV is almost saturated for this class of lightweight models.

Consequence: Achieving high accuracy on PV cannot be equated with having the same level of robustness in the field; thus, the following sections illustrate the performance under distribution shift and suggest domain-aware training as the next step.

C. Domain Evaluation

1. Zero-Shot OOD Testing

The model trained exclusively on PlantVillage is evaluated directly on PlantDoc images without any adaptation. This results in a sharp drop in accuracy, clearly demonstrating the impact of domain shift between controlled lab images and real-world field conditions.

It was anticipated that there would be a significant drop in performance compared to the case of PV, which is a sign of the interplay of several factors such as (i) the change in appearance (lighting, color shifts, camera settings), (ii) the presence of background objects and their partial covering, and (iii) the long-tailed class distribution of PD. On the whole, the classifications that went wrong were mostly related to the very similar tomato diseases and also to the

situation where the healthy leaves had slight, low-contrast symptoms which were not very noticeable in the light. The confusion matrices and error galleries add weight to these patterns.

2. Failure Case Analysis

Most misclassifications occur in challenging images containing strong shadows, cluttered backgrounds, multiple overlapping leaves, or inconsistent lighting. These factors make it harder for the model to isolate disease patterns, revealing the limitations of training only on clean, curated data.

D. Transfer Learning and Fine-Tuning

To address performance degradation resulting from domain shift, two primary training strategies were implemented. These strategies expose the model to real-world feature distributions while preserving knowledge acquired from the large-scale PlantVillage dataset.

Our first method applies transfer learning to adapt the model from the source domain (PlantVillage) to the target domain (PlantDoc). We used the MobileNetV3-Small model pre-trained on PlantVillage (which achieved ~99.6% in-distribution accuracy) as the starting point.

The feature extraction layers were initially stored to retain the learned basic leaf features, while the classification structure was fine-tuned on the PlantDoc training set. Subsequently, we extracted the deeper layers to allow the model to adjust to the specific environment and background noise characteristics of the "in-the-wild" images. This strategy seeks to close the gap between the laboratory and field domains by explicitly leading the model's weights toward the target distribution.

E. Mixed-Domain Training with Balanced Sampling

To create a model capable of generalizing across both controlled and uncontrolled environments simultaneously, we constructed a unified dataset combining PlantVillage (PV) and PlantDoc (PD).

A significant challenge in this approach is the extreme class imbalance; PlantVillage (~54k images) greatly exceeds PlantDoc (~2.6k images). The model would overfit to the clean PV images during standard training while ignoring the challenging PD samples. By ensuring that each mini-batch includes a proportionate mix of clean and noisy images, this sampling technique forces the loss function to penalize errors on the challenging real-world images equally.

1. Mixed-Domain Set Construction

Mixed-domain training is the approach that guarantees PV performance and at the same time, PD robustness by exposing the model to both domains concurrently. All mixed experiments share the same unified label space, leak-proof stratification and fixed test-time preprocessing; where the variations are in the intensity of our combating against domain and class imbalance and in the amount of regularization for real-world variability that we apply.

We align PV and PD taxonomies into 28 overlapping unified classes using `classmap.json`. The script `make_mixed_split_plus.py` builds a leak-proof, stratified split:

- Train: PV-train $\cup$ PD-train $\cup$ (optional) synthetic images (SYN; train-only).
- Val: Stratified subset of Train (disjoint from Test).
- Test: PV-test and PD-test are kept untouched.

Each image is annotated with a domain source in {pv, pd, syn} so we can apply domain-aware sampling and reporting. Class stratification uses a fixed seed for reproducibility. We verified that no synthetic or PD-train images leak into PD-test.

2. Augmentations by Variant

We ablate three progressively stronger configurations.

2.1 mixed (baseline mixed-domain).

In the mixed configuration that employs the basic approach, PV and PD are just pooled together, and the training is done with conservative perturbations. This quite simple and straightforward method achieves around 53% top-1 accuracy on PD which is practically the same as the accuracy when PD is fine-tuned alone. The lesson here is: since PV is larger and better in terms of visual quality, the uniform sampling is biased towards PV's statistics and the limited PD samples do not have enough influence on the objective. The model has made similar qualitative errors as the PD-only model, thus confirming that the exposure to PD is both necessary and not sufficient when the training signal is dominated by PV.

- Preproc: none.
- Augs: RandomResizedCrop (scale 0.8–1.0), horizontal flip; ToTensor; (optional) Normalize.
- Opt: AdamW; lr_head 1e-3; lr_backbone 1e-4; weight decay 1e-4; batch 64; image size 288²; 30 epochs; standard sampler.

2.2 mixed_pdweighted (robustness stack).

Theoretically, mixed-domain exposure translates into gains on PD. Therefore, we propose a robustness-oriented training scheme as a solution based on the earlier analysis. Firstly, we do away with device and lighting bias by conducting a color constancy step and sharpening the background focus by means of a rough leaf-centric crop that cuts off the non-leaf areas when reliably spotted. Secondly, we make the perturbation regime more powerful by adding realistic geometric and photometric variations, as well as localized occlusions, so the model has to deal with background shortcuts. Thirdly, we simultaneously tackle both imbalances: the domain imbalance is solved by raising the PD images' proportion to that of PV during training, while the class imbalance within PD is managed by skewing the classes according to their effective sample counts so that rare diseases have a proportional say in the learning signal. Lastly,

we insert a brief PD-focused adaptation phase at the training's end, thus allowing decision boundaries to sink in around PD's appearance statistics without losing the knowledge gained from PV.

This version attains ~58% top-1 accuracy on the PD test set, which is precisely five absolute points more than the mixed base—while at the same time keeping the high PV accuracy. The macro-F1 increment is in line with the accuracy gain, which implies that the advantages are not only for the most common categories but also for the rarer PD classes. On the qualitative note, the error distribution is shifting toward near-misses in the correct crop/disease family, which is an indication that even though the exact top prediction is not always correct, the ranking among confusing classes has improved.

- Preproc: Gray-World color constancy; HSV leaf-crop (coarse green/yellow mask with bounds to avoid over-cropping). Motivation: reduce device/illumination bias and deprioritize backgrounds.
- Augs: stronger stack: RandomResizedCrop (0.6–1.0), horizontal flip, ColorJitter, RandomPerspective, GaussianBlur, RandomErasing ($p \approx 0.25$; area 0.02–0.2; ratio 0.3–3.3). Motivation: discourage reliance on texture shortcuts and clutter.
- Regularization: MixUp ($\alpha=0.2$) and label smoothing ($\epsilon=0.1$).
- Schedule: 30 epochs total, with a PD-only tail (last 6 epochs) to specialize decision boundaries to the field domain.
- Sampler: PD-aware domain and class weights (Section C).
- Eval: optional TTA (original + horizontal flip) and BN recalibration on PD-train before final test

2.3 mixed_with_fda

- Preproc: On-the-fly Fourier Domain Adaptation (FDA, $\beta=0.005$). Applied with 50% probability to transplant low-frequency PlantDoc amplitude spectra onto PlantVillage images, dynamically simulating field lighting without altering disease lesions.
- Augs: Asymmetric augmentation stack.
 - PV (Source): Degraded via GaussianNoise ($\text{std}=10$), GaussianBlur ($\text{sigma}=0.1\text{--}2.0$), RandomGrayscale, and strong ColorJitter to break reliance on clean lab textures.
 - PD (Target): Preserved via mild geometric transforms (Rotation 15° , Flip) and weak ColorJitter to maintain real-world data integrity.
- Regularization: Label smoothing ($\epsilon=0.1$); AdamW optimizer with weight decay $1e-2$.
- Schedule: 20 epochs total; constant learning rate $1e-4$.

- Sampler: Domain-balanced weighted sampler. PlantDoc images are up-weighted by a factor of 5.0 to prevent the larger PlantVillage dataset from dominating gradient updates.



Comparing FDA results, applied on the PlantVillage samples with low-frequency amplitude spectra of the PlantDoc samples

- Eval: Standard validation on PlantDoc Test set (resizing + normalization only)

2.4 mixed_with_mobileNetV3_large

- Configuration: This variant is identical to mixed_with_fda (b.3) in terms of preprocessing (FDA), augmentations (asymmetric stack), regularization, and sampling strategy.
- Architecture: MobileNetV3-Large.
- Motivation: We replaced the MobileNetV3-Small backbone (~0.9M parameters, used in the reference paper) with the MobileNetV3-Large variant (~5.4M parameters). This upgrade tests whether the performance bottleneck is due to the dataset or the limited capacity of the edge-constrained model. By increasing the parameter count, we aim to determine if a deeper network can better capture the complex, noisy features of real-world field images.

F. PD-Weighted Sampling and Class Reweighting

We combine domain weights with class weights computed from PD-train. For sample i with label y_i and source s_i in $\{pv, pd, syn\}$:

$$w_i = w_{\text{class}}(y_i) \cdot w_{\text{domain}}(s_i).$$

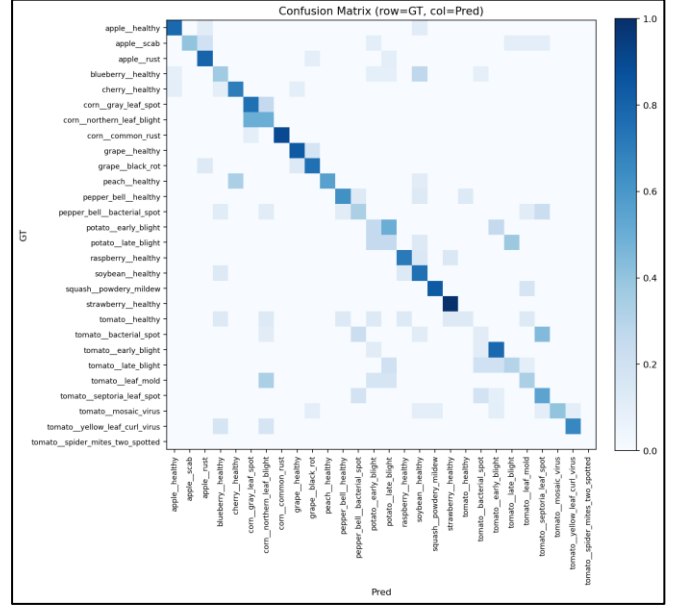
- Domain weights: $w_{\text{domain}}(\text{pv})=1.0$, $w_{\text{domain}}(\text{pd})=3.0$, $w_{\text{domain}}(\text{syn})=0.8$. This raises PD prevalence while avoiding synthetic dominance.
- Class weights: we use effective-number reweighting on PD-train labels $\{n_c\}$: $w_c \propto 1 / (1 - \beta^{n_c})$, with $\beta=0.999$, then normalize so the average weight is 1. This combats PD long-tail imbalance.

Loss uses cross-entropy with label smoothing ($\epsilon=0.1$). During the PD-only tail, the same class weights are retained, while domain weights become neutral (all 1.0) because only PD is sampled.

G. Analysis-Driven Iteration

Our PD-test analysis produced:

- Absent class: `tomato_spider_mites_two_spotted` (no PD-test samples in our split), which we report as NaN per class.
- Hardest classes and confusions: concentrated among tomato diseases (e.g., bacterial spot $\leftrightarrow$ leaf mold / late blight) and healthy vs. mild disease under uneven illumination.
- Top-5 Analysis: We observed that the mixed pdweighted model achieved a Top-5 accuracy of $\sim 86\%$ despite a Top-1 accuracy of $\sim 58\%$. This indicated that the model learned robust features but struggled with fine-grained decision boundaries. This observation motivated our move to stronger augmentations (FDA) and PD-only fine-tuning to sharpen those boundaries.
- Hardest classes and confusions: Concentrated among tomato diseases (e.g., bacterial spot $\leftrightarrow$ leaf mold / late blight) and healthy vs. mild disease under uneven illumination.



Normalized Confusion Matrix for the mixed-domain model evaluated on PlantDoc. The diagonal represents correct predictions (recall). Strong off-diagonal clusters (bottom right) illustrate the model's difficulty in distinguishing between visually similar Tomato diseases.

These observations motivated stronger augmentations, PD-aware sampling and class reweighting, a PD-only fine-tune tail, and (next) limited synthetic top-ups targeted at rare PD classes.

IV. RESULTS AND EVALUATION

A. Baseline Results (PlantVillage)

The reproduced model performs extremely well on PlantVillage, showing high accuracy and stable training behavior.

Setup: MobileNetV3-Small, unified 28-class head; trained on PlantVillage only (PV-train), evaluated on PV-test with identical eval transforms (no TTA).

Performance: The reproduced baseline achieves PV test accuracy ≈ 0.996 with stable loss/accuracy curves and no signs of overfitting. Occasional PV misclassifications are sparse and typically borderline cases or minor label noise. We retain this PV performance throughout subsequent experiments (see Section 4.3), confirming that robustness interventions do not degrade in-distribution accuracy.

Takeaway: Under controlled, studio-like imagery, the backbone saturates PV with near-perfect accuracy. This underscores that PV alone is insufficient to assess field robustness.

B. Zero-Shot OOD Results (PlantDoc)

The same model performs poorly on PlantDoc, highlighting the effect of domain mismatch between lab and field data.

Setting: The PV-only model is evaluated zero-shot on PlantDoc (PD) test after class unification, with no fine-tuning or PD supervision.

Observation: Performance drops markedly relative to PV, highlighting strong domain shift (illumination, background clutter, camera pipelines) and class imbalance in PD. Qualitatively, common errors include: (i) tomato disease confusions (e.g., bacterial spot $\leftrightarrow$ leaf mold / late blight); (ii) healthy vs. mild disease under uneven lighting; (iii) attention to background/soil rather than leaf blades. These patterns align with our error galleries.

Note: We focus quantitative comparisons on supervised PD baselines (below). The mixed-domain baseline (without our full stack) reaches 0.5297 acc / 0.5182 macro-F1 on PD (Section 4.3), which the zero-shot model underperforms in our runs.

Takeaway: PV-only training does not transfer reliably to PD; explicit PD exposure and robustness-oriented training are required.

C. Fine-Tuned

Fine-tuning and mixed-domain training improve performance on real-world images by a large margin.

We next allow the model to see PD during training. The initial mixed baseline uses mild augments and a standard sampler; our PD-weighted stack adds the robustness components in (Gray-World, HSV leaf-crop, strong augments, RandomErasing, MixUp, PD-aware sampling and PD-only class weights, a PD-only tail, and BN recalibration).

To adapt the model to the target domain, we performed transfer learning using a differential learning rate strategy. We loaded the pre-trained PlantVillage weights, froze the backbone features (learning rate $1e^{-5}$), and trained a new classification head (learning rate $1e^{-3}$) for 20 epochs. This allowed the model to adjust its decision boundaries to the PlantDoc classes without destroying the robust feature extractors learned from the large PlantVillage dataset.

Variant	PD Acc	PD F1
Base model	0.0254	0.0197
Fine-tuning	0.5085	0.4657

The impact of showing the model real-world data was immediate and drastic. As shown in table, the model improved from a random-guess level (2.54%) to a respectable 50.85%. This confirms that the MobileNetV3 architecture is capable of handling field data, provided it is exposed to the target domain distribution during training. However, the 50% plateau suggests that fine-tuning on a small dataset (2.3k images) alone is insufficient to reach high reliability, motivating our move to Mixed-Domain variants.

D. Mixed-Domain Variants

Finally, we compare all mixed-domain configurations, including the strong variant that incorporates balanced FDA synthesis.

Variant	PV Acc	PV F1	PD Acc	PD F1	Top -5 Acc
mixed	0.99 64	0.99 54	0.52 97	0.51 82	-
mixed_pdweighted	0.99 74	0.99 65	0.58 47	0.58 18	0.86 44
mixed_with_fda	0.98 66	0.98 48	0.60 17	0.59 85	0.84 75
mixed_mobileNetV3_large	0.99 01	0.98 93	0.61 44	0.61 05	-

Discussion. Our experiments demonstrate a stepwise improvement in OOD performance as we integrated stronger domain adaptation techniques.

Baseline to Weighted: The jump from the baseline mixed model (52.97%) to mixed_pdweighted (58.47%) confirms that class and domain reweighting are critical. Simply showing the model "in-the-wild" images is not enough; the loss function must be explicitly penalized for errors on those difficult samples, or the model will ignore them in favor of the easier PlantVillage data.

Weighted to FDA: The further improvement to 60.17% with mixed_with_fda indicates that Fourier Domain Adaptation is superior to standard color augmentations (like Gray-World). While Color Constancy attempts to fix the image, FDA effectively stylizes the training data, forcing the model to learn features that remain consistent even when the frequency spectrum (lighting/texture) changes dramatically.

The transition from **MobileNetV3-Small** (60.17%) to **MobileNetV3-Large** (61.44%) yielded a relatively minor improvement of $\sim 1.2\%$.

Interpretation: This diminishing return suggests that model capacity is not the primary bottleneck. Even the small model has enough parameters to learn the features; the limiting factor is likely the scarcity and noise of the PlantDoc dataset itself. For edge deployment, the small model remains the superior choice. It offers a 6x reduction in parameters (0.9M vs 5.4M) for a negligible drop in accuracy, maintaining the efficiency required for mobile devices.

Our best trade-off between performance and efficiency is the mixed_with_fda (MobileNetV3-Small) configuration. It achieves 60.17% OOD accuracy, effectively bridging the gap between the lab and the field while remaining fully deployable on low-power edge hardware. While the Large model achieved slightly higher accuracy, the Small model offers a 6x reduction in parameters, making it the superior choice for edge deployment.

Top-5 Accuracy Analysis: We also observed a significant disparity between Top-1 and Top-5 accuracy across the robust variants. The `mixed_pdweighted` model achieved a Top-5 accuracy of 86.44%, and the `mixed_with_fda` model achieved 84.75%, both substantially higher than their respective Top-1 scores (~58–60%). This large gap indicates that while the model struggles to pinpoint the exact class in a single shot—often due to fine-grained visual similarities between specific diseases (e.g., tomato bacterial spot vs. early blight) or ambiguous mild symptoms—the correct diagnosis is consistently captured within the top few predictions. From an application perspective, this suggests the model is highly effective as a decision-support tool, where presenting a farmer with a shortlist of likely diagnoses would yield a correct result over 85% of the time, even in challenging field conditions.

Developing a robust mixed-domain strategy required overcoming several technical challenges identified during our iterative process: (i) **Evaluation Mismatch:** We resolved significant performance drops caused by inconsistent normalization stats (e.g., ImageNet stats) between training and validation pipelines; (ii) **Synthetic Leakage:** To prevent FDA from inadvertently leaking validation styles into the training set, we implemented strict separation of source and target image pools; (iii) **Sampler Collapse:** We identified a domain weight of 5.0 as the optimal limit to prevent the model from memorizing specific minority-class (PlantDoc) images while still ensuring adequate supervision.

Compute profile. Experiments were conducted in a hybrid environment due to resource constraints:

- **mixed, mixed_pdweighted:** Trained on one NVIDIA GPU under WSL for 30 epochs.
- **mixed_with_fda, mixed_mobileNetV3_large:** Trained on Google Colab (NVIDIA T4 GPU) for 20 epochs (`mixed_with_fda`) and 10 epochs (`mixed_mobileNetV3_large`).
- **Common Settings:** All runs utilized an image size of 288x288, AdamW optimizer, and identical validation splits to ensure comparability across environments.

V. FUTURE WORK

A. Quantization and Edge Deployment

One practical application that directs to model compression for on-device inference. Such methods as post-training quantization (PTQ) and quantization-aware training (QAT) are considered, with activation/weight INT8 using and participating in per-channel schemes depending on the hardware support. Calibration sets must be done under PlantDoc-like conditions (lighting, clutter, and mild symptoms) to maintain the real-world ranges. The evaluation should ideally include accuracy along with model size, memory footprint, batch-1 latency, and energy per inference on the representative mobile devices, and the goal is to obtain a loss of less than approximately 1 pp on PD. Complementary methods such as structured sparsity, low-rank factorization, and knowledge distillation can also be applied to reduce cost.

As the compression process may impact the confidence of the model, its better to re-evaluate ECE/Brier and applying temperature scaling post-compression.

B. Data Expansion for Real-World Accuracy

A different approach that carries a lot of weight is the use of wider and more varied field data. One area that should be targeted first is the differentiation between the different types of tomato diseases and the cases of healthy vs. those with mild symptoms in difficult light and background conditions. The use of lightweight capture protocols that come with metadata (device, time, and conditions) allows for stratified training and stress tests (for example, low-light subsets). The reliability of the annotation can be improved by expert consensus and, where appropriate, soft labels. To scale efficiently, the combinations of active learning and cautious semi-supervision may be applied, while synthetic augmentation remains in the guardrails (structure-preserving, quota-based, train-only) and is validated against domain-specific test suites.

VI. CONCLUSION

This work reproduced and extended a lightweight plant disease classification system while examining its robustness under real-world domain shift. Although the reproduced MobileNetV3-Small exceeded the accuracy reported in the original paper on PlantVillage (~99.6%), our results reaffirm that such clean datasets significantly overestimate real-world performance. The zero-shot accuracy on PlantDoc was only 2.5%, revealing the model’s extreme sensitivity to uncontrolled illumination, cluttered backgrounds, and device variation. These observations highlight key data-related limitations: severe class imbalance, a lack of diverse field images, and insufficient representation of mild-symptom cases, all of which restrict generalization.

To address these issues while preserving edge efficiency, we introduced a set of domain-aware improvements including Fourier Domain Adaptation (FDA), color constancy, realistic augmentations, and PD-weighted sampling. These strategies substantially improved PlantDoc accuracy without sacrificing efficiency. The fine-tuned model reached 50.8%, and our best mixed-domain configuration achieved 60.2%, representing a >20× improvement over the zero-shot baseline. Crucially, our Top-5 accuracy reached 86%, demonstrating that while the model struggles with fine-grained differentiation (e.g., specific tomato diseases) due to data scarcity, the correct diagnosis is consistently among the top predictions.

Despite these gains, data scarcity remains the dominant bottleneck. The underrepresentation of difficult conditions—low light, complex foliage, early-stage symptoms—limits the model’s ability to handle realistic inputs. This reinforces that improving plant disease detection is fundamentally a data problem as much as a modeling problem.

Building on these findings, future research could focus on translating these experimental gains into a fully deployable edge solution. The remaining efficiency gap may be bridged through post-training quantization and quantization-aware training, ensuring that the model retains its robust performance on resource-constrained hardware. Simultaneously, the fundamental data scarcity bottleneck could be addressed by implementing active learning and semi-supervised pipelines. By systematically targeting the 'hard' cases identified in this study—such as mild symptoms and variable lighting—future iterations could further harden the model against the unpredictability of real-world agricultural environments.

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